

Project Workflow: Week 2

Phase: Requirements Analysis & System Design

1. Objective

The primary goal of Week 2 was to define the technical requirements and organizational structure of the Block Blast game. During this phase, the gameplay mechanics, user interactions, scoring system, and software architecture were carefully analyzed to establish a clear foundation before implementation. This planning process helped reduce future development issues and ensured that every module would work together efficiently.

2. Detailed Task Breakdown

- Requirement Finalization: Identified the functional and non-functional requirements of the game, including block movement, collision detection, scoring rules, and game-over conditions.
- System Organization: Designed the modular architecture by separating the Game Controller, Game Board, Block Manager, Score System, and UI Manager.
- Interface Planning: Prepared the initial user interface layout consisting of the game board, score display, next-block preview, and control buttons.
- Design Review: Evaluated the planned workflow to ensure scalability, maintainability, and efficient future development.

3. Deliverables

- Functional Requirements Specification
- System Architecture Design
- User Interface Wireframe
- Module Relationship Diagram

4. Tools & Resources

- Java Development Kit (JDK)
- IntelliJ IDEA / NetBeans IDE
- Draw.io / Figma
- Git & GitHub
- Microsoft Word for Documentation

Core System Modules

Module Name	Primary Responsibility
Game Controller	Manages the main game loop and application state.
Game Board	Maintains the playing grid and block placement.
Block Manager	Generates and controls different block shapes.
Score System	Calculates and updates player scores.
UI Manager	Handles menus, buttons, and player interactions.